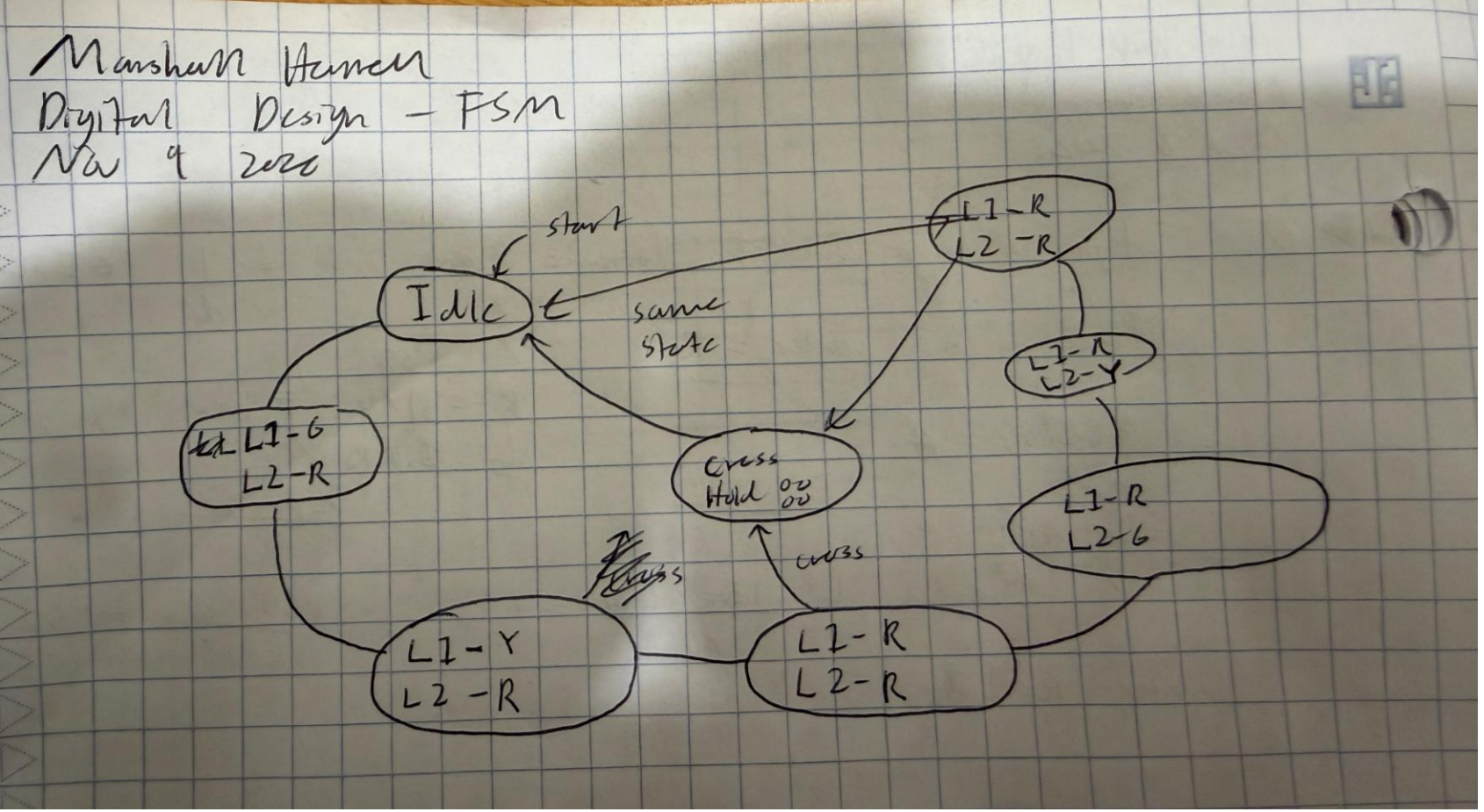
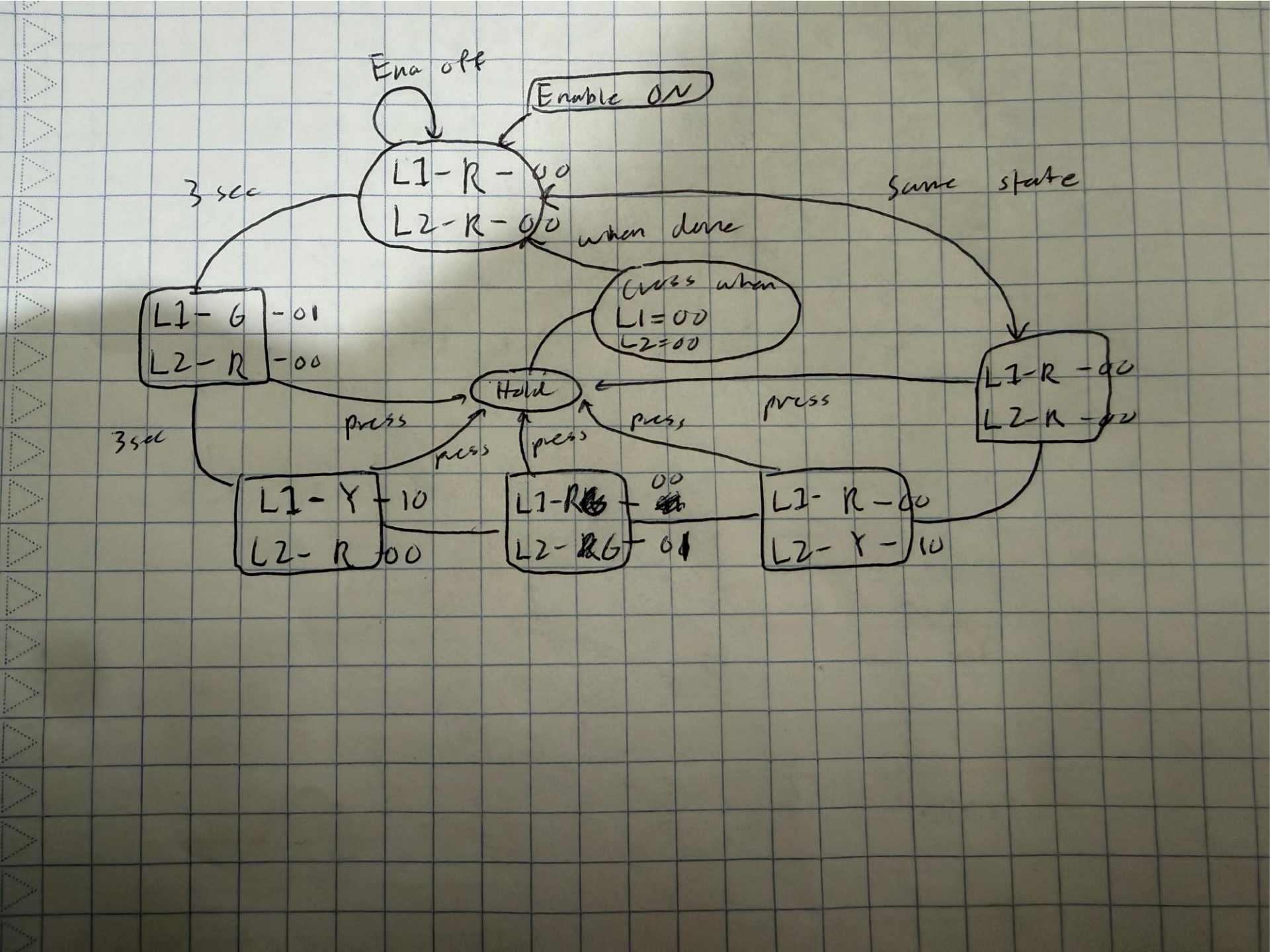
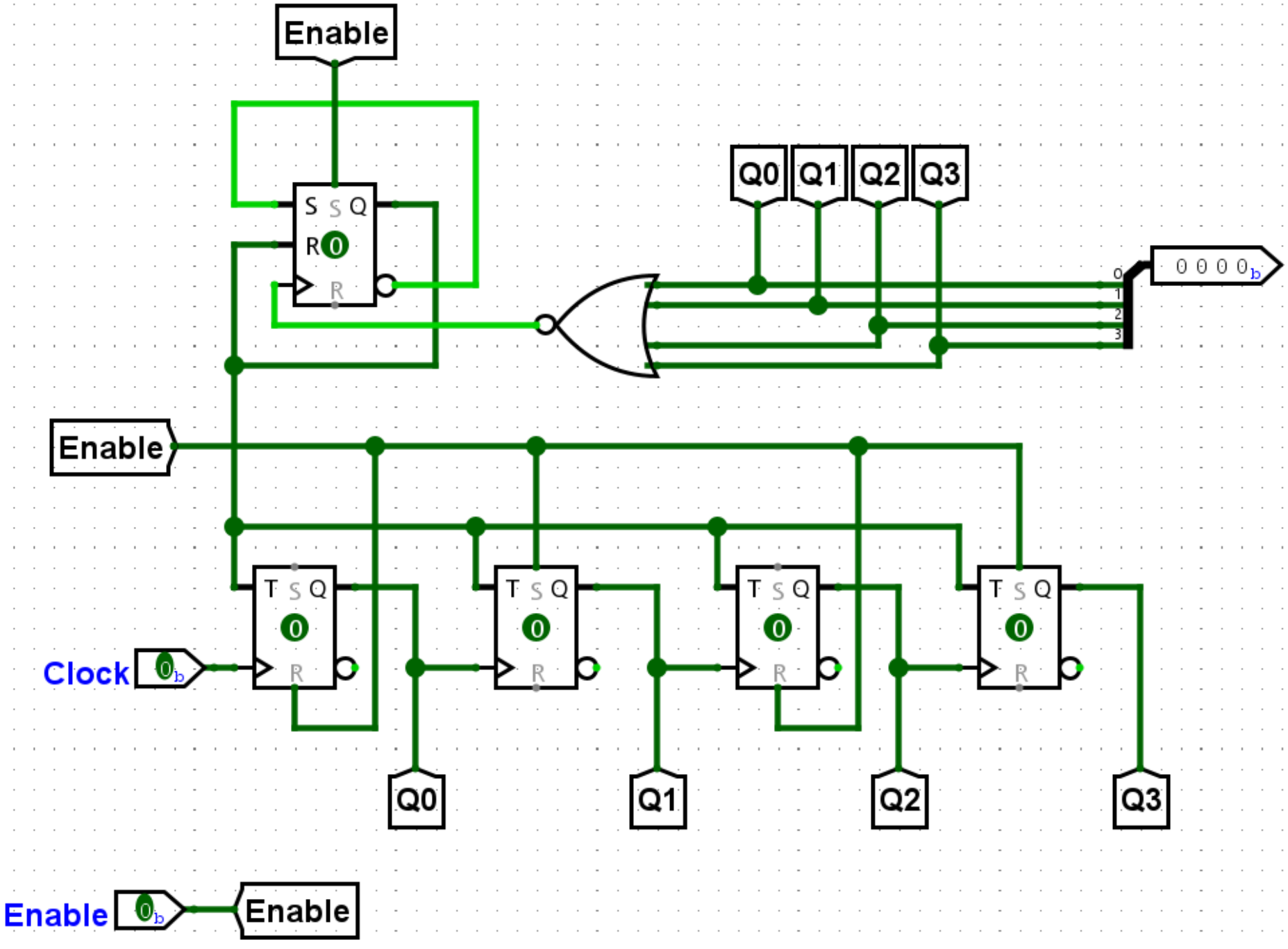


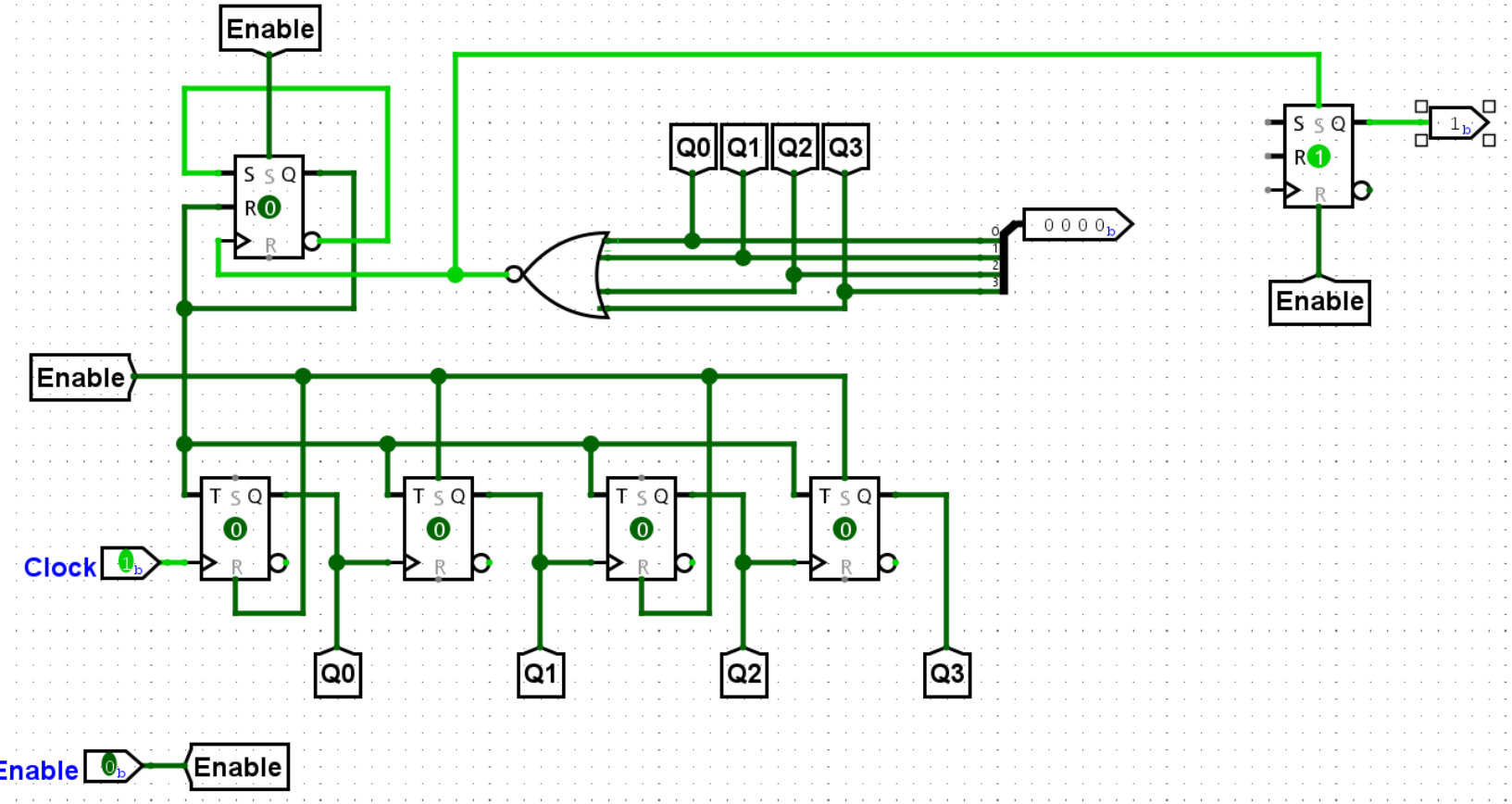
- First FSM. Did not really account for holding the value when the button was pressed adnd waiting for red



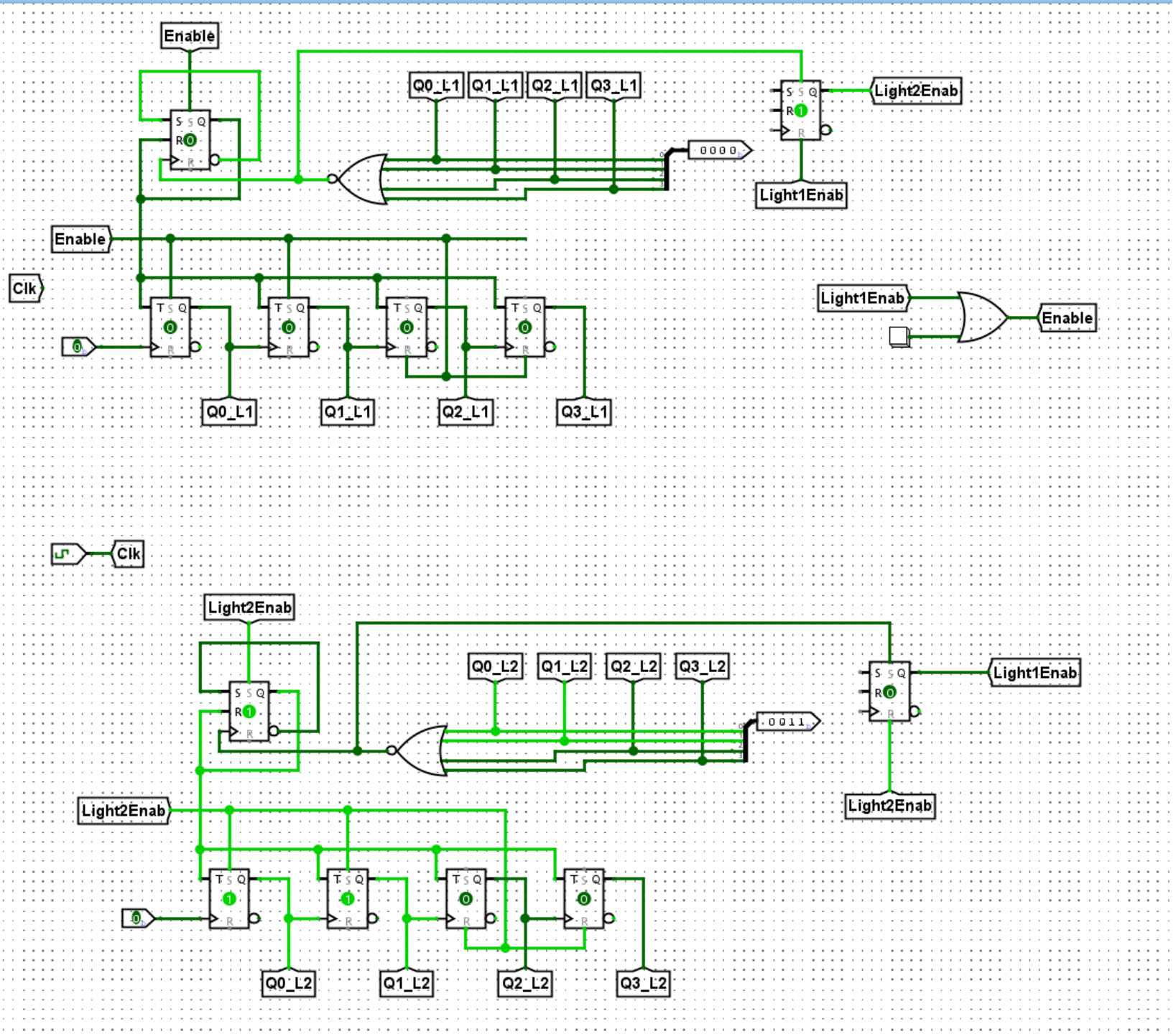
- Pressing Button holds the value until both light states are red.
- Enable must be on for the entire system to work
- Rotates through lights normally until button is pressed

First design for the down counter

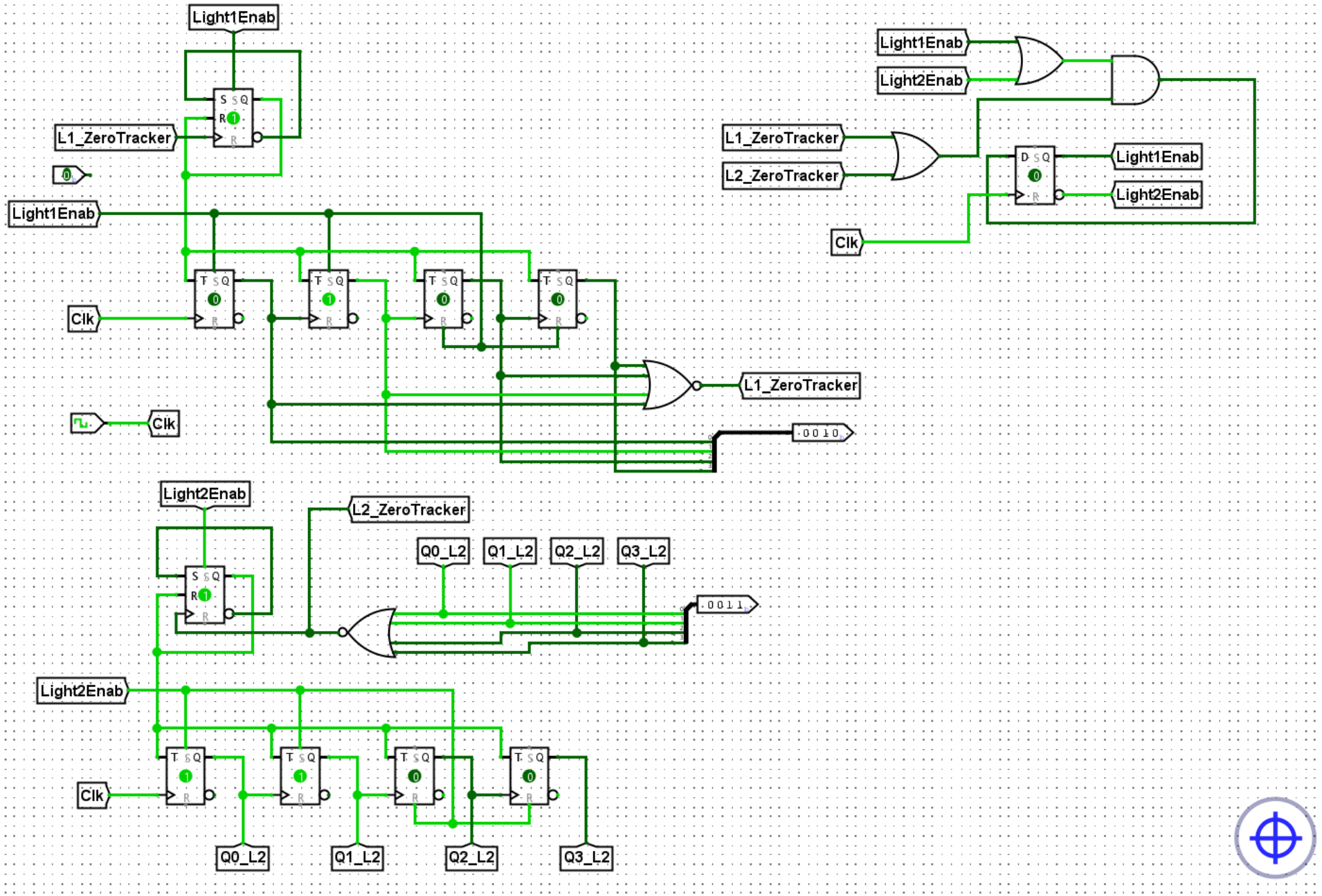




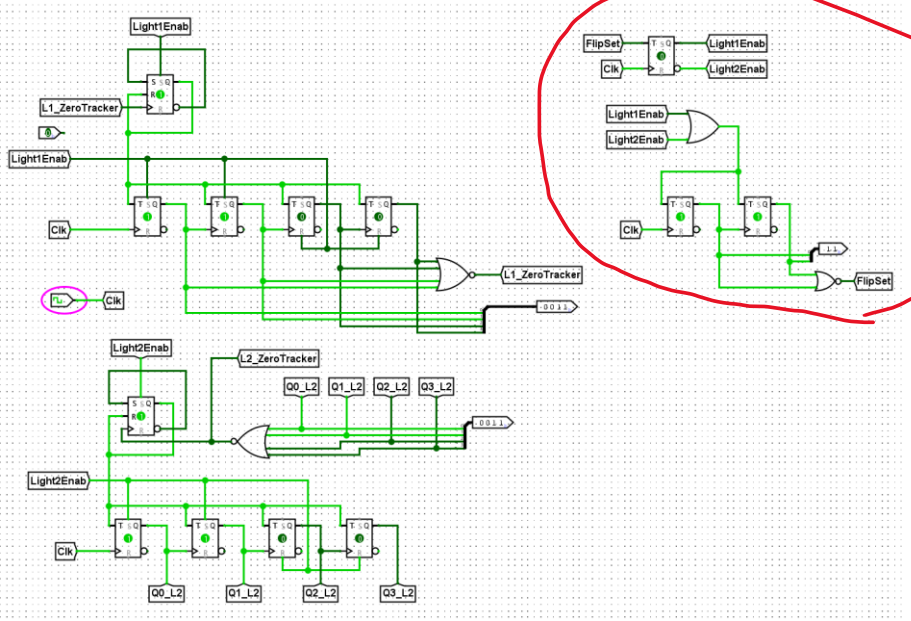
- Trying to add a latch to track which state light is going and which one is staying red

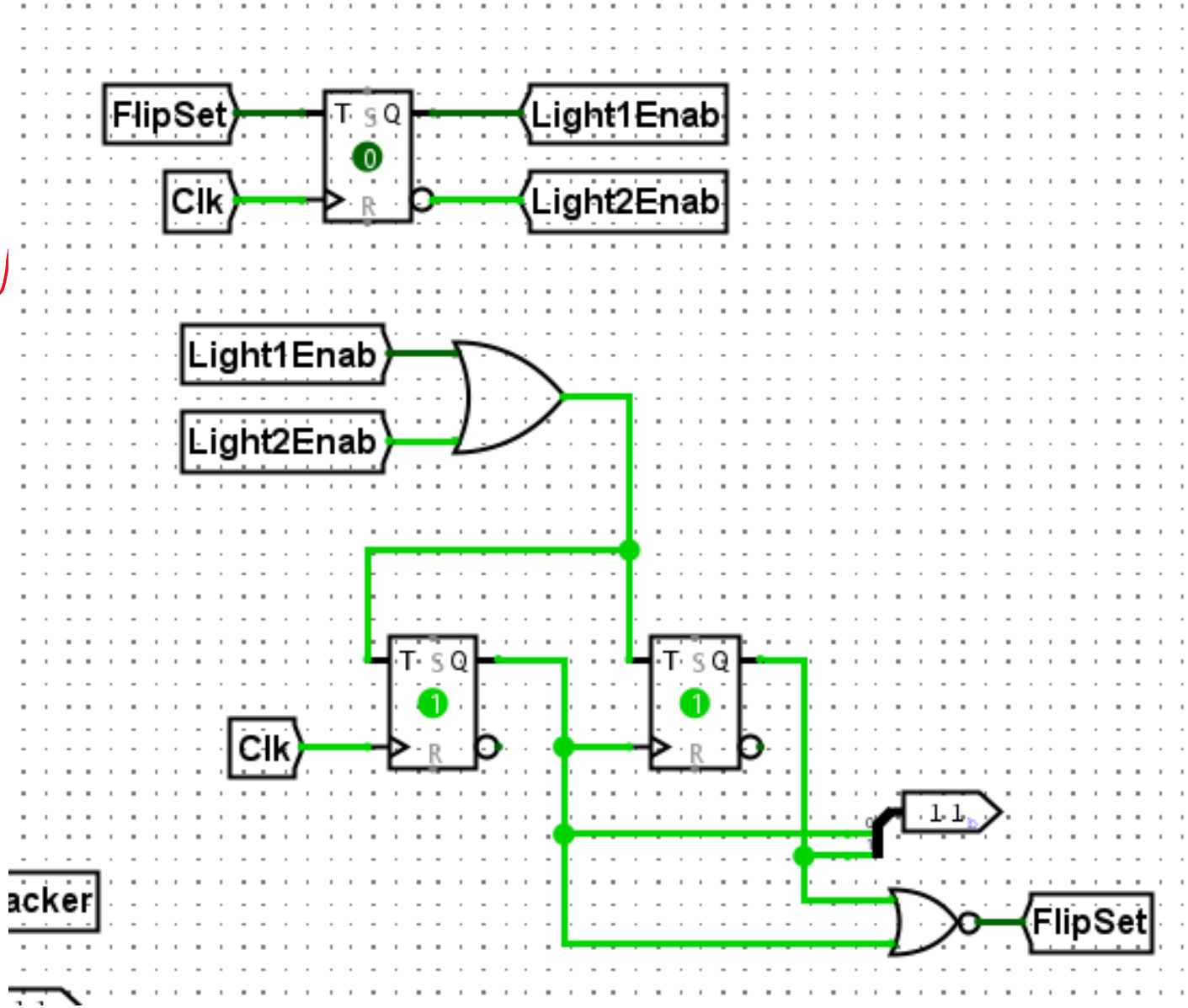


- Make the lights flip flop themselves. Once one finishes, the next one starts.
- Only used button for now. Going to try and use the flip flop idea bur for the whole system. The end of one causes the start of the other one.

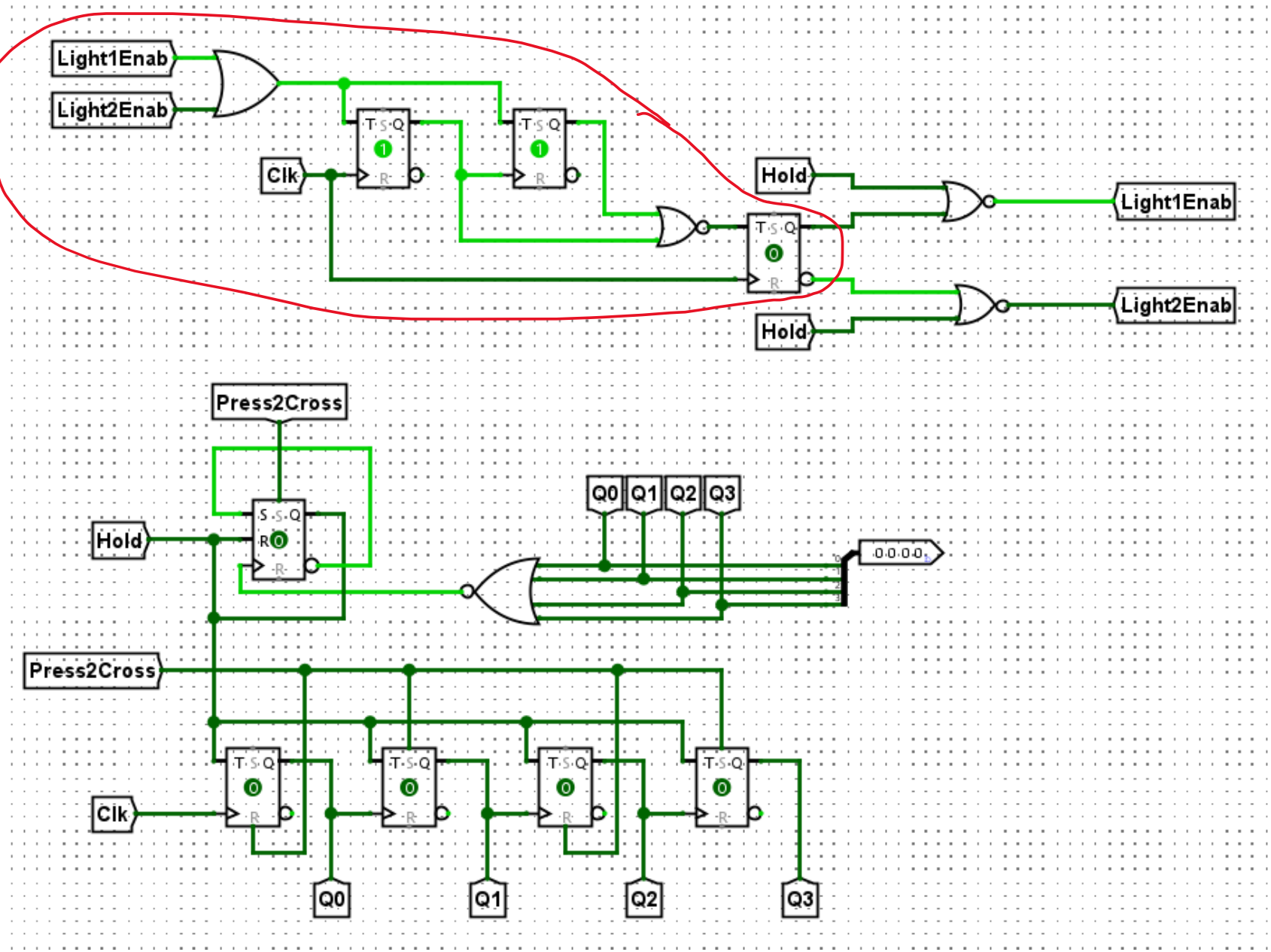


- This was my original idea for how to get the 2 states to flip back and forth after 3 seconds
- This correctly held one at the red state and the other count down, and when it reached zero flipped to allow the other one to count down, but not for 3 seconds

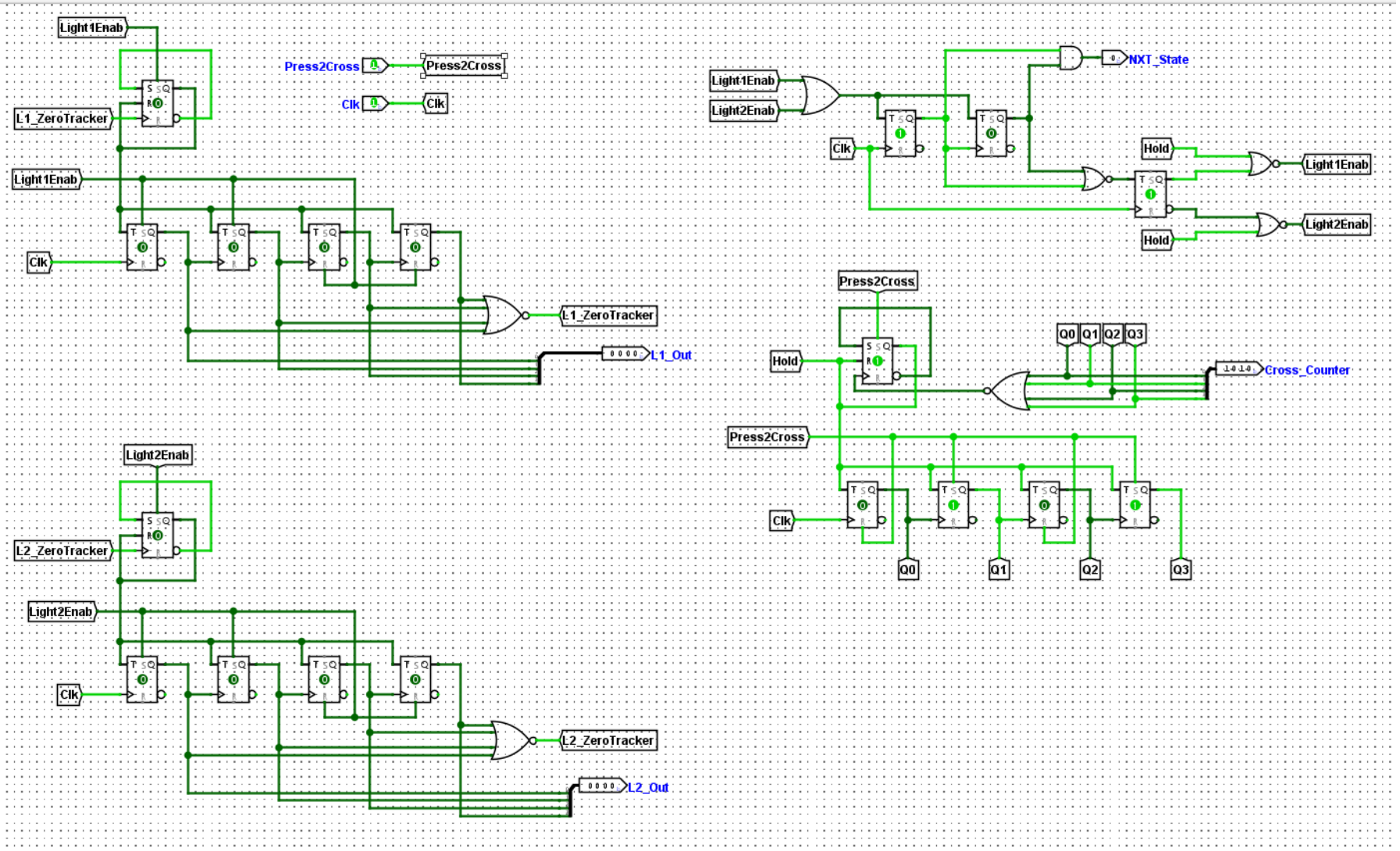




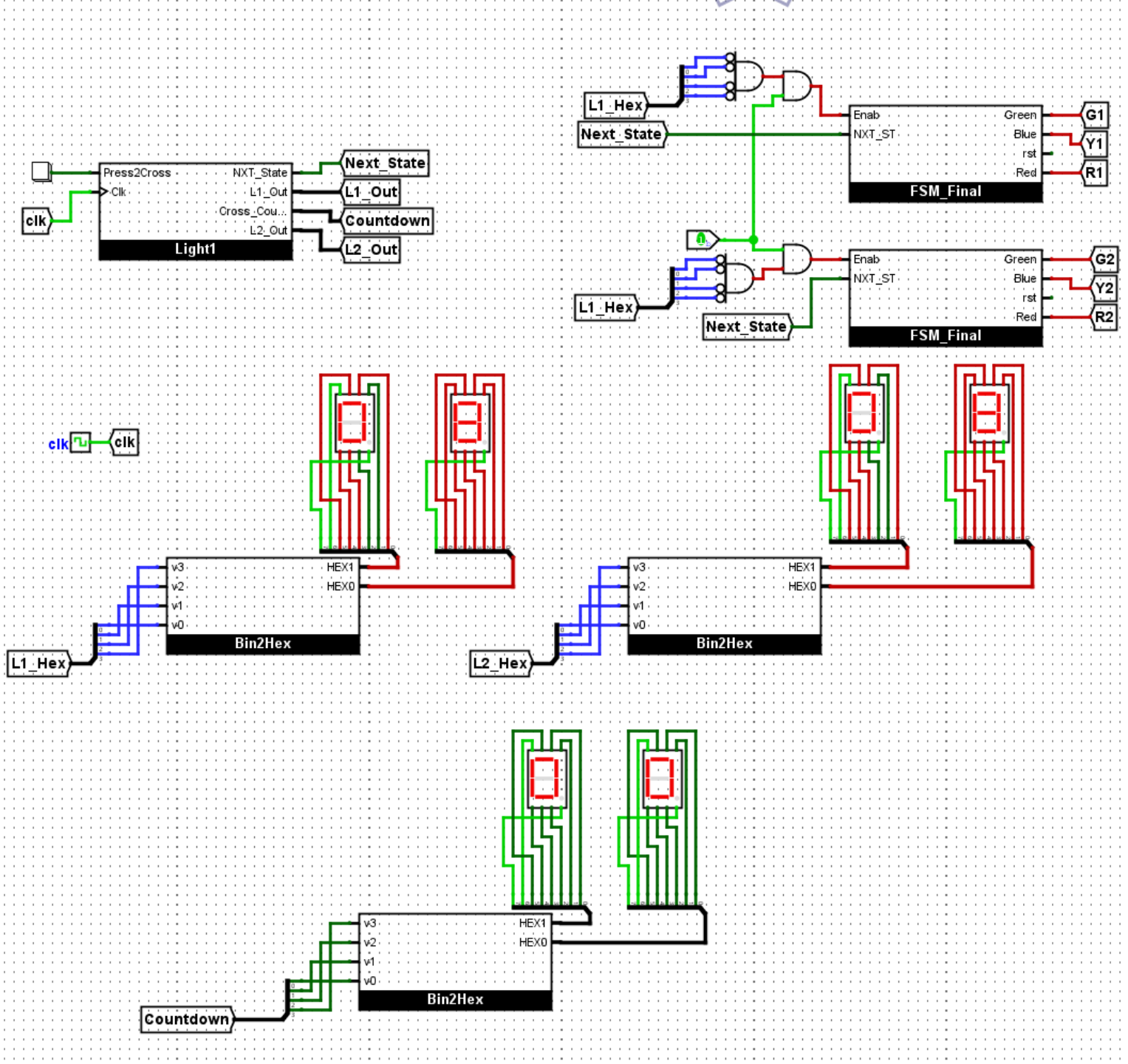
- The design circles allows counts two 3 with the others, and when it reaches three, utilizes a T Flip-Flop to allows the other to hold, and the other to count down.
- This makes me think that holding the other enable on as means to not allow it to countdown won't be helpful for the traffic light ☹



- The red circle does the same functions as the last one (rewired a little bit)
- Added the 10 second down counter that only is utilized when a button is pressed (pedestrian)
- Added a “hold” button to the enable pins. The Hold is only on when the cross-counter is in use. That is connected to nor gates that pause the light count downs



- This is the farthest point that I could manage to get to before getting too stuck before submitting.
- The circuit goes through the 3 light states for one light and when its done triggers the second light to start.
- Button pauses both lights and runs the 10 second cross counter



- Idea for what final solution might look like. I think that my FSM tab is messed up or doesn't cooperate with my circuits as I need it to.
- Need to figure out how to get the counters to not have the red errors.